From Last Place to Group Gains: Low-Performers and Group Spillovers*

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Abstract

How do performance improvements of *laggards*, the lowest-performing individuals in a team, influence group performance? Drawing on social comparison theory and the concept of last place aversion, we suggest that the improvement of a laggard's performance can serve as a catalyst for enhancing the entire group's output, because other group members will enhance their own efforts to avoid falling into last place. Using nearly eight years of data from an industrial firm, we examine how laggards' performance influences the performance of their groups. Our results show that improvements in laggards' performance significantly enhance subsequent performance of the remaining group. This performance enhancing effect is particularly pronounced for those group members ranked near the bottom of the hierarchy. Moreover, this effect differs between groups and depends on how salient it is within groups that the laggard can change and how close the performance of other group members is to the laggards' performance. Our findings extend research on rank-based motivation by documenting field evidence consistent with last place aversion and its contingent impact on group performance.

Keywords: Group performance, last place aversion, downward comparison

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1 Introduction

A majority of Fortune 500 companies rely on performance comparison mechanisms in their incentive programs, such as performance dashboards or ranking schemes (Kwoh, 2012; Alavi et al., 2025). These programs typically, and often quite rationally, focus heavily on rewarding top performers, or those positioned near the top of the ranking hierarchy. For instance, IBM's recognition of top sales representatives improved and stimulated greater effort among other group members, resulting in a significant rise in overall group performance (Kiron, 2012; Aguinis and Bradley, 2015).

The importance of high performers for overall group success goes beyond the direct contribution of their own performance. A large body of empirical research has found that lower performers engage in upward comparisons with higher performers, motivating them to increase their own effort and performance (Gubler et al., 2016; Obloj and Zenger, 2017; Ahearne et al., 2025). It is therefore unsurprising that most existing research and practitioner activity is overwhelmingly focused on those either at or towards the top of the ranking. Practically speaking, managers primarily aim to facilitate competition for the top end of the rankings, and through this aim to drive overall group performance.

This understandable focus on top performers neglects another potential mechanism for driving group performance, namely *last place aversion* (Kuziemko et al., 2014). Last place aversion denotes an individual's strong desire to avoid feeling that they are at the bottom of a hierarchy, expressed through an individual's desire to escape their own occupation of last place in a ranking, or to avoid falling into last place. We term those individuals in last place as laggards. Being the laggard in a ranking or competition is frequently associated with failure and incompetence (Wiltermuth and Flynn, 2013; Ferris et al., 2015; Buell, 2021), and may be perceived as particularly undesirable by individuals, seeing the elevated salience and attention associated with the lowest ranking place.

Importantly, prior research on employee performance rankings has solely focused on individuals at the last place (Song et al., 2018; Gill et al., 2019; Boichuk et al., 2019), and their attempts to move out of that position. In this research we propose, and empirically demonstrate, that aversion to last place extends to those employees who are close to last place in a performance ranking, thereby showing that performance improvements of those at the bottom of the ranking have spillover effects on the performance of other group members, and thus substantial effects on overall group performance.

Evidence from other rank-based contexts suggests that aversion to last place is both psychologically strong and behaviorally consequential. For example, individuals are willing to switch waiting queues to avoid being in last place, even if this decision results in a suboptimal outcome (Buell, 2021). Similarly, individuals express strong dissatisfaction with social policy changes, such as welfare redistribution, when these changes may cause them to fall into last place in income distribution or social hierarchy (Kuziemko et al., 2014; Xie et al., 2017; Martinangeli and Windsteiger, 2021), even if such changes would leave them better off in absolute terms. However, while this literature shows that there is a strong aversion to falling into last place among individuals in various social situations, it does not effectively inform how this aversion unfolds in groups within organizations. If last place aversion arises in group settings, improvements in a laggard's performance may prompt other group members to intensify their own effort in order to avoid slipping into last place, thereby increasing overall group performance. This would equip managers with an alternative and previously underexplored pathway for influencing group outcomes, rather than focusing only on top or high-level performers.

Therefore, the key objective of our study is to explore whether increases in the performance of laggards increase overall group performance (over and above the effect of their own contribution). Drawing on the idea of last place aversion (Kuziemko et al., 2014) and social comparison theory (Festinger, 1954), we argue that as the laggard's performance improves, it may motivate their peers in the group to enhance their performance, so as to avoid falling into last place themselves. In this way, the laggard's performance improvement can act as a catalyst for enhanced effort among other group members and, in turn, elevate overall group performance. Furthermore, we explore whether this effect depends on whether the group member occupying the last place changed since the previous month, the magnitude of the performance difference between the laggard and the penultimate group member, and the magnitude of the performance difference between the laggard and average group performance.

To explore these possibilities, we acquired a longitudinal dataset from an industrial organization, capturing sales performance from 1,279 employees from 35 groups across 7 years and 11 months. The groups in this organization consist of groups of salespeople who are regularly informed about their performance and associated rank changes within their group, allowing us to observe the relationship between the laggard's performance and the performance of their peers. To investigate how changes in laggards' performance relate to the subsequent performance of their groups, we use monthly performance data and analyze it using an instrumental variable approach while accounting for unobserved heterogeneity.

Our results confirm a significant effect of laggard performance improvements on the subsequent performance of the rest of the group. This positive effect is stronger for group members positioned closest to the bottom of the ranking, who face the most immediate risk of falling into last place. Moreover, we empirically confirm the theorized contingencies. Specifically, the positive impact of a laggard's performance improvement is amplified when the individual who was ranked as the laggard recently changed, signaling greater instability in bottom-rank positions. In contrast, the impact weakens as the performance gap between the laggard and the second-to-last employee grows, and also as the overall performance disparity between the laggard and the rest of the group increases.

Our study makes three significant contributions to theory on the drivers of employee performance, and on performance rankings. Specifically, we are the first to provide empirical evidence that individuals increase their own performance in response to increases in the performance of the laggard in their group, plausibly to avoid falling into last place. We thus transfer the phenomenon of last place aversion, which has been observed in a general societal context (Kuziemko et al., 2014), to the important context of performance ranking in organizations. While existing research in the workplace has explored the motivation of laggards themselves to escape that specific position (Gill et al., 2019; Boichuk et al., 2019), we show that other group members' aversion to falling into last place can also motivate them to improve their performance and increase group performance. Further, we are the first to identify positive effects of last place aversion on those near last place. For example, prior work on social welfare suggests that aversion to last place results in irrational resistance to positive welfare policies (Kuziemko et al., 2014). In contrast, we show that last place aversion can act as a positive motivator, encouraging peers of the laggard to improve their own performance, ultimately benefiting their own, and the group's, outcomes.

Finally, our study advances understanding of the heterogeneity inherent in last place aversion. Specifically, we show that the response to laggard performance improvements depends on both the shift of occupancy of the last-ranked position, and the size of the performance gap separating the laggard from proximal peers. We extend the theoretical arguments of LePine and Van Dyne (2001) and Dutcher et al. (2015) on comparative performance and effort provision by empirically demonstrating how these rank dynamics shape the intensity of last place aversion and drive peer performance within real organizational groups.

Our findings have several significant implications for managerial practice. Most prominently, managers should attempt to leverage the often significant, untapped performance potential of laggards. Given diminishing marginal returns to investment, investing in laggards, as compared to incremental investments in top performers, might yield substantial productivity improvements and returns. For example, while Boichuk et al. (2019) suggest that removing laggards can be a potential practice in managing laggards, their approach raises an unresolved question: when a laggard is removed, the next lowest-performing individual inevitably moves into last place, creating a new laggard. Thus, at what point should the manager stop removing laggards? Instead, our findings suggest an approach that exploits last place status (regardless of the absolute performance level of the laggard) to lift up the overall group: Showcasing the progress of laggards and their efforts to catch up with peers can boost group performance by motivating continuous improvement, particularly among group members whose performance is close to the bottom rank.

2 Related Literature

2.1 Performance comparison and ranking

Our study explores performance comparisons between group members and the individual occupying the lowest rank in the group, which are enabled by the use of visible performance rankings within the group. Theoretical foundations for performance comparison are grounded in social comparison theory (Festinger, 1954), which posits that individuals routinely evaluate their performance by comparing themselves to others. This process serves as a mechanism for self-evaluation and helps individuals make sense of their performance within a social context (Buunk and Gibbons, 2007).

Prior research (see Table B1 in the Appendix for representative empirical studies) has examined the effects of various aspects of performance comparisons on outcomes such as individual productivity (Obloj and Zenger, 2017), competitive behaviors (Garcia and Tor, 2007; Vriend et al., 2016), and performance sensitivity (Gartenberg and Wulf, 2017). Existing studies of the effects of social comparison via rankings typically focus on constrained comparison contexts, most commonly involving dyadic or narrowly defined rank contrasts, such as comparisons with a single adjacent competitor or with a fixed reference point. However, performance comparisons within groups in organizations unfold within a dense social environment characterized by ongoing interaction (Lim et al., 2009). In such settings, individuals are embedded in groups where rank positions are continuously recalibrated, and performance comparisons are not necessarily limited to a single peer but may occur relative to multiple or all peers within one's group (Ahearne et al., 2025).

Importantly, existing literature on rank-based comparison is heavily weighted towards the effects on lower performers of their upward comparisons with higher-ranking performers (Garcia et al., 2013; Charness et al., 2014; Vriend et al., 2016), and horizontal comparisons with peers (Gartenberg and Wulf, 2017). In contrast, downward comparisons, those directed toward lower-ranking individuals, have received comparatively little attention, particularly in organizational performance settings. Although downward comparisons and their consequences have been acknowledged in broader social comparison research (Buunk and Gibbons, 2007), their role in shaping performance within performance rankings in organizations remains unexplored.

The present study addresses these gaps by shifting attention to downward comparisons and incorporating the ranking of the entire group into the analysis of performance comparisons. We examine how group members respond to changes in the performance of the lowest-ranked individual and how this response varies across positions in the ranking. Additionally, we explore contingencies derived from group-level ranking dynamics that may moderate the effects of performance comparisons. In doing so, we connect downward comparison dynamics within group hierarchies to the broader concept of last place aversion, which we elaborate in the following section.

2.2 Last Place Aversion

While upward comparisons have traditionally dominated the performance comparison literature, recent research suggests that individuals respond particularly strongly to the prospect of occupying the bottom of a ranking, a position frequently associated with social stigma, failure, and diminished status (Martinangeli and Windsteiger, 2021). This behavioral tendency was initially conceptualized in the context of individuals close to the bottom of the socioeconomic income distribution (i.e. those in the second-to-last place), and referred to as *last place aversion* (Kuziemko et al., 2014). Management and organizational researchers have explored a similar aversion to last place, but from the perspective of the individual actually ranked last. Such studies have found that individuals ranked last experience a strong desire to escape their position, which influences their effort provision (Gill et al., 2019), rational choice (Buell, 2021), and productivity (Song et al., 2018). Table B2 in the Appendix provides a summary of representative studies on last place aversion.

Further, existing research on last-place aversion largely focuses on either individuals who currently occupy the bottom rank, treating this position as exogenous (Gill et al., 2019; Buell, 2021), or those who occupy the second-to-bottom rank (Kuziemko et al., 2014). These perspectives, however, overlook broader group-level dynamics, and the possibility that last-place aversion may extend beyond individuals at or immediately above the bottom of the ranking. Given that individuals are strongly averse not only to being last but also to the prospect of falling into last place (Kuziemko et al., 2014), in group performance settings, a laggard's effort to escape the bottom rank may therefore alter multiple other members' perceived risk of downward rank mobility. Observing improvements by the lowest-ranked member can increase the perceived threat of being overtaken, motivating those in various positions above the laggard to increase effort to protect their relative standing. Because such spillover effects of last-place dynamics on higher-ranked members remain almost entirely unexplored, existing literature does not yet offer a comprehensive account of how last-place aversion shapes both individual and group performance.

Our study therefore extends the literature on last place aversion by considering how changes in the laggard's performance influence broader group-level outcomes, through their impact on multiple non-laggard individuals within a group ranking. By linking the interactions between the laggard and their peers in a group context, we address the limitations of prior studies that focus primarily on static last place effects.

2.3 Summary

Building on these two streams of literature, we aim to understand how downward performance comparisons with the laggard of a group influence the performance of other group members. Unlike upward comparisons, which have been extensively discussed in existing social comparison studies (Garcia and Tor, 2007; Obloj and Zenger, 2017; Waltré et al., 2023), downward comparisons in general have received very little attention in the incentives, rankings, and performance literature. However, management research has shown that individuals have a distinct fear of being in last place (Gill et al., 2019), and socioeconomic studies suggest that falling into last place is also significantly aversive (Kuziemko et al., 2014). Bringing these two streams of research together suggests that the specific downward comparison context of the aversion to last place might serve as a powerful motivator, driving individuals to either escape the bottom rank, or to avoid falling into it. We integrate this phenomenon into the context of within-group comparisons, focusing on the interactions between laggards and their group peers.

3 Last-place aversion and peer effort

The prior discussion provides a qualitative account of effort provision under last-place aversion. We now formalize the mechanism through which an improvement by the lowest-ranked group member, *the laggard*, changes the effort incentives of other group members. The model yields our central proposition and provides the theoretical basis for our main hypothesis that laggard performance improvements enhance peer effort and consequently, increase subsequent group performance

Consider in a group, a non-laggard member i chooses effort $e_i \geq 0$ before performance outcomes are realized. The member's performance is:

$$y_i = \alpha_i + \theta e_i + \varepsilon_i, \tag{1}$$

where α_i captures member-specific ability, $\theta > 0$ is the productivity of effort, and ε_i is an idiosyncratic performance shock. We assume that $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$.

Let y_L denote the observed performance of the laggard who occupied last place in the preceding period.[1] Because group members observe performance rankings, y_L provides a benchmark for the lower boundary of the group's performance distribution. A higher value of y_L means that the performance of the laggard has moved upward, increasing the exposure of other group members to the possibility of falling into last place.

Last-place exposure. For a non-laggard member i, we define exposure to the bottom rank as the probability that the member's current performance falls below the prior-period laggard's observed performance:

$$P_i^L(e_i; y_L) = \Pr(y_i < y_L) = \Phi\left(\frac{y_L - \alpha_i - \theta e_i}{\sigma}\right), \tag{2}$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. This probability increases as the laggard's performance y_L rises and decreases as group member i exerts more effort. We define z_i as:

$$z_i = \frac{y_L - \alpha_i - \theta e_i}{\sigma}. \tag{3}$$

The term z_i captures the standardized distance between the laggard's performance and group member i's expected performance. For a non-laggard member whose expected performance exceeds the laggard's performance, $z_i < 0$. When a group member i's expected performance is close to laggard's, values of z_i closer to zero, meaning that the risk of falling into last place is more salient. When a member's expected performance is far above the laggard's, value of z_i will be more negative, and the perceived probability of falling into last place is small.

Expected utility. The group member's realized payoff is

$$\pi_i = by_i - \frac{\kappa}{2}e_i^2 - -\psi\,\mathbf{1}\{y_i < y_L\}, \tag{4}$$

where $b > 0$ captures the monetary benefit of performance, $\kappa > 0$ governs the cost of effort,

[1] In our empirical setting, individuals regularly observe the performance and rank of every group member through weekly and monthly performance review meetings, as described in Section 4.1. We therefore treat prior-period laggard performance as an observable benchmark when current effort choices are made.

and $\psi > 0$ is the utility loss associated with falling below the laggard's performance. We distinguish last-place aversion from a standard model of effort provision: the group member values effort not only because it increases expected performance, but also because it reduces exposure to the bottom rank.

Substituting (1) and (2) into the utility function and taking expectations before the performance shock is realized gives:

$$U_i(e_i; y_L) = b(\alpha_i + \theta e_i) - \frac{\kappa}{2}e_i^2 - \psi\Phi\left(\frac{y_L - \alpha_i - \theta e_i}{\sigma}\right), \tag{5}$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function. The final term is the expected disutility associated with falling below the laggard's performance.

For the group member, they chooses effort to maximize equation (5). At an interior solution, optimal effort e_i^* satisfies the following, where $\phi(\cdot)$ is the standard normal density function.

$$\kappa e_i^* = b\theta + \frac{\psi\theta}{\sigma}\phi\left(\frac{y_L - \alpha_i - \theta e_i^*}{\sigma}\right), \qquad \text{let } z_i^* = \frac{y_L - \alpha_i - \theta e_i^*}{\sigma} \tag{6}$$

Equation (6) separates two motives for effort. $b\theta$ is the standard marginal benefit of effort from increasing expected performance. The second term is the additional marginal benefit of effort generated by last-place aversion. Effort lowers the probability that the individual falls below the laggard's performance, and this effect becomes more important when the individual faces a risk of doing so.

Consider a non-laggard group member whose expected performance exceeds the laggard's performance $(z_i^* < 0)$. An increase in the laggard's performance raises the member's optimal effort. Formally,

$$\frac{\partial e_i^*}{\partial y_L} = \frac{-\psi\theta z_i^*\phi(z_i^*)/\sigma^2}{\kappa - \psi\theta^2 z_i^*\phi(z_i^*)/\sigma^2} > 0 \qquad \text{for } z_i^* < 0. \tag{7}$$

When the laggard improves, the bottom-performance benchmark rises, leaving other group members with a smaller buffer between their expected performance and the bottom rank. In that case, the perceived probability of falling into last place is high. This increases the perceived risk of falling into last place, especially for group members whose performance is close to the laggard's. Because effort reduces this probability, laggard improvement increases the marginal value of effort and induces other group members to work harder. The proof of the model is provided in Appendix A.

4 Empirical setting and Hypotheses

4.1 Empirical setting

To emprically investigate this phenomenon, we require an empirical setting with three conditions. First, the study requires group-based environments where individuals operate as part of a group but exhibit varying levels of individual performance. This ensures the performance heterogeneity required for performance comparisons. Second, as we focus on rank-based performance comparisons, it is critical that group members are not only ranked based on their individual performance but also have visibility of these rankings. This transparency allows individuals to reflect on their relative standing within the group and experience performance comparisons (Ahearne et al., 2025). Third, we require a context in which performance is measurable and assumes an important role as an explicit goal in the organization.

Based on these criteria, we collaborated with a large industrial firm that markets construction supplies and products (e.g., fastening materials, tools, chemicals) to industrial clients across Europe. The company maintains a substantial sales department, comprising nearly a thousand salespeople, whose primary duties involve selling products and providing consultation to their customers. As is common in industrial selling, the company organizes its sales force in regional units. Each unit includes a manager as the group leader and between 3 and 19 salespeople. Once assigned to a group, salespeople remain in that group and do not switch. For each group, they are responsible for a distinct geographic region or client portfolio, ensuring that there is no overlap in market coverage across groups. Within a given group, salespeople perform comparable roles and operate under the same organizational structure and incentive system. Each salesperson is individually responsible for managing their own customers and prospects, and there is no sharing of accounts or sales responsibilities within the group.

Importantly, each group is required to hold weekly sales catch-up meetings and monthly group performance reviews. This systematic approach keeps every salesperson informed about every group member's sales performance and performance ranking. For instance, if the bottom-ranked salesperson improves their performance, other group members quickly become aware of this change. This level of transparency is common in business-to-business sales forces (Steward et al., 2010; Chen and Chung, 2021; Ahearne et al., 2025).

4.2 Hypotheses

Our main hypothesis concerns the relationship between a laggard's performance and the performance of the rest of the laggard's group members. We draw on social comparison theory (Festinger, 1954; Buunk et al., 1990) and last place aversion (Kuziemko et al., 2014). Social comparison theory suggests that individuals make sense of their performance by comparing themselves to others. We focus on the relatively unexplored phenomenon of downward comparisons, and specifically where salespeople evaluate their standing relative to the last place group member. In this context, laggards occupy a salient position within groups, making their performance a focal point of comparison for others (LePine and Van Dyne, 2001).

In line with last place aversion (Kuziemko et al., 2014), we suggest that other group members compare their performance with the laggard to avoid falling into last place themselves. In these comparisons, increases in a laggard's performance signal a threat to other members in the group, increasing the perceived risk of being overtaken and falling into last place. As individuals are strongly averse to occupying the lowest rank (Kuziemko et al., 2014), the progress of the laggard should motivate other group members to increase their effort, leading to higher group performance. We therefore expect the following:

H1: An improvement in the group laggard's performance positively affects group performance.

In addition, we expect that the effect of laggard performance on group performance is heterogeneous and depends on the likelihood of the laggard catching and passing other group members in the ranking. If this likelihood increases, other group members should invest greater effort to avoid falling into last place themselves (Garcia et al., 2013; Kuziemko et al., 2014), increasing their performance in response to the laggards' performance improvement. Drawing on situational factors driving social comparison and competitive behavior (Garcia et al., 2013), we consider three group characteristics influencing this likelihood and the intensity of last place aversion effects. In groups where the laggard has changed recently, last place aversion should be more salient among other group members. Further, in groups where the performance differences between the laggard and the salesperson ranked second-to-last, or the performance differences between the laggard and the rest of the group, are smaller, last place aversion should be stronger as there is greater proximity to the last place.

First, we expect that changes in the identity of the laggard increase the effect of laggard performance improvement on group performance. Garcia et al. (2013) identify social category boundaries as key situational factors that intensify social comparison by making categorical distinctions between group members more salient. As the laggard occupies a salient position within groups, group members will recognize when the individual in this position changes (LePine and Van Dyne, 2001) . Such a change indicates to other group members that falling into last place is a genuine possibility and that group ranks are volatile. This uncertainty should prompt increased social comparison (Festinger, 1954) and thus strengthen last place aversion effects among group members, leading to greater effort and associated performance increases.

Furthermore, the likelihood of other group members becoming the laggard should be influenced by their proximity to the laggard. Proximity increases comparison concerns and competitiveness because the closer individuals are to a performance threshold, the more intensely they respond to social comparison (Garcia et al., 2013). In group rankings, being the laggard reflects a performance threshold that group members want to avoid (Kuziemko et al., 2014; Boichuk et al., 2019). Proximity in group performance rankings can be captured in two ways. The first is the magnitude of the performance difference between in the laggard and the second-to-last ranked group member. The second is the group-level performance disparity between the laggard's performance and the performance of other group members. Both differences should indicate how much safe ground there is between the laggard and other group members. The smaller this safe ground is, the more likely other group members are to become the laggard, thus increasing their last place aversion. As a result, in groups with smaller differences in performance between the laggard and the second-to-last ranked group member, and groups with smaller disparity between the laggard and the group-level performance, the effect of a laggard's performance increases on group performance should be more pronounced. Drawing on the above logic, as last place aversion increases (decreases) due to variation in these three factors, we expect the following:

H2a. A change in the identity of the laggard within a group strengthens the positive relationship between laggard performance and group performance.

H2b. A larger performance difference between the laggard and the second-lowest-performing group member weakens the positive relationship between laggard performance and group performance.

H2c. A larger performance disparity between the laggard and the other group members weakens the positive relationship between laggard performance and group performance.

5 Data and Method

5.1 Data source

We obtained objective sales performance data from the company's Enterprise Resource Planning (ERP) system, which provides unique identifiers for both salespeople and their corresponding groups, and monthly observations on several performance metrics including revenue, number of active selling days, and tenure records. The data include 1,279 salespeople across 7 years and 11 months, for a total of 39,772 monthly observations of sales performance. All salespeople in the data set are formally employed by the firm, and remained with the firm for at least six months. The salespeople in our sample are distributed across 35 sales teams. On average, each group consists of 12 salespeople.

We conduct our analysis in two stages. In the first stage, our primary objective is to determine how laggard performance affects overall group performance. In the second stage, we explore heterogeneity in this effect. Table 1 presents operationalizations of all variables that we use in our analysis.

5.2 Key Variables

The key dependent variable for our main analysis is group performance. We measure group performance as the average sales revenue that individuals within each sales team i generated in month t. To calculate the average group revenue, we exclude the observations for laggard salespeople within each group. In our additional analysis, we focus on individual performance as our dependent variable, measured as the revenue each salesperson generates per month.

Our independent variable is laggard performance. We measure laggard performance as the sales revenue that a laggard in each sales team i generated in month $t-1$ and thus one month before observing group performance. Table 2 reports correlations and descriptive statistics for variables included in our main analysis.

Table 1: Variable Operationalizations

Dependent variable	**Operationalization**
Group performance	Mean monthly group sales revenue excluding the sales revenue of the last-placed salesperson in the month, calculated as (total sales revenue of the group – laggard's sales revenue) / (number of salespeople in the group – 1).
Key predictor and instrument	
Laggard performance	The sales revenue generated by the lowest-ranked salesperson in a group in a given month.
Laggard active selling days	The total count of unique days within a given month during which the laggard generated sales revenue. This variable serves as an instrumental variable in our analyses.
Active selling days of group peers	The median count of unique days within a given month during which the group peers of a laggard generated sales revenue. This variable serves as a control variable in our IV specification.
Moderators	
Laggard position change	A binary variable that indicates whether the individual occupying the last-place position within a group changed since the previous month.
Performance gap	The difference between the lowest and second-to-lowest sales revenue generated by salespeople in a group in a given month.
Magnitude of performance disparity	The degree to which laggard performance deviates from that of other group members. It is calculated using the root mean square formula (see Equation 4).
Variables used in additional analysis	
Performance of penultimate person	The second-lowest sales revenue generated by a salesperson in a group in a given month.
Performance of median person	The median sales revenue generated by a salesperson in a group in a given month.
Performance of top person	The highest sales revenue generated by a salesperson in a group in a given month.

Table 2: Correlation and Descriptive Statistics of Key Variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) ln(Group performance)								
(2) ln(Group performance$_{t-1}$)	0.79**							
(3) Laggard performance$_{t-1}$	0.14**	0.23**						
(4) Laggard active selling days$_{t-1}$	0.08**	0.17**	0.79**					
(5) Active selling days of group peers$_{t-1}$	0.21**	0.37**	0.17**	0.16**				
(6) Laggard position change	0.07**	0.08**	−0.06**	−0.10**	0.02			
(7) Performance gap	0.12**	0.23**	−0.15**	−0.13**	0.16**	0.12**		
(8) Magnitude of performance disparity	0.18**	0.49**	−0.16**	−0.15**	0.49**	0.05*	0.38**	
Mean	20,467.00	20,382.20	2,656.80	8.70	19.60	0.40	2,882.30	22,013.00
Standard deviation	8,427.40	8,387.80	3,475.10	6.40	2.10	0.50	2,801.10	9,573.00

Notes: * $p < 0.05$; ** $p < 0.01$. Group performance and group performance at $t-1$ are log-transformed in the correlation matrix, consistent with how they enter the empirical model. Means and standard deviations are based on the raw measures of the original variables.

5.3 Identification strategy

Lagged responses of last-place aversion effects. Because group members become aware of a laggard's performance change within their group through frequent (weekly and monthly) team meetings, their own performance adjustments likely materialize after a short time lag only. For instance, when noticing that an underperforming peer is improving their performance, other salespeople may invest more time and effort in selling additional products or acquiring new customers to safeguard their own standing, and avoid falling into last place. Consequently, the effects of a laggard's performance improvement on other group members' performance could in principle be observable as early as the following month, given that salespeople are often quick to adapt their strategies and update their sales activities (Spiro and Weitz, 1990; Franke and Park, 2006; Alavi et al., 2019). Therefore, we lag our independent variable, laggard performance.

Unobserved heterogeneity across groups. Groups may possess unobserved characteristics that impact their daily operations and, subsequently, salesperson performance. These unobserved factors could range from the customer relationship and sales territory of a group to other factors like leadership style and group cohesion. In the absence of controlling for these factors, there is a risk of omitted variable bias if these unobserved characteristics drive both laggard performance and group performance. For instance, a group with an exceptional leader might create an environment where both the laggard and the rest of the group can excel, leading to a correlation between the two performance outcomes. To address this issue, we include group fixed effects in our model to control for all time-invariant heterogeneity within each group.

Seasonality and common time trends. Sales performance may be affected by unobserved factors that change over time but remain consistent across all salespeople, such as seasonal variations in customer demand or organization-wide changes, such as the adoption of new technology (Peers et al., 2012; Soysal and Krishnamurthi, 2012). Omitting these factors from the model introduces the risk of bias into our model. To account for seasonal and unobserved common time trends, we incorporate monthly time fixed effects into our model.

Group-specific demand fluctuations. Despite controlling for group and monthly fixed effects, the relationship between laggard performance and group performance may still be confounded by group-specific demand fluctuations. For example, maybe due to regional industry policies and economic trends, or temporary changes in customer demand, both laggards and other group members experience similar increases or decreases in their performance. This could result in a spurious correlation between laggard performance and group performance. To address this concern, we apply a two-stage least squares (2SLS) approach. We use *laggard active selling days* as an instrument to control for endogeneity in laggard performance. The variable captures the extensive margin of the laggard's selling activity by measuring the number of distinct days in a month during which the laggard was engaged in revenue-generating activity. Laggard active selling days increase when the laggard devotes greater effort to selling.

The validity of our instrument hinges on its relevance and exogeneity. In terms of relevance, the primary question is whether our instrument can sufficiently explain the variation in laggard performance. In our institutional context, salespeople commonly generate sales revenue when they sell products to customers. Thus, conditional on other determinants of sales revenue, a laggard who records more active selling days is more likely to achieve higher monthly revenue. On the contrary, any interruptions to selling activity, such as personal leave or illness, are likely to result in a corresponding reduction in their revenue, as reflected by the reduced active selling days in the relevant month. As we describe further down below, the results of our first-stage estimation support this argument.

Importantly, the exclusion restriction assumption requires that our instrument does not affect group performance via pathways other than through laggard performance, given that we control for group heterogeneity and common time trends. Our institutional context helps support this assumption. Each salesperson is solely responsible for the orders from their assigned customers, and customer accounts are not shared within groups. Thus, an additional laggard active selling day does not directly generate sales orders or customer interactions for other group members.

To further reduce the concern that laggard active selling days might be correlated with group-level demand conditions, we include the *active selling days of group peers* as an additional control variable in both stages of our instrumental variable specification. This variable captures the contemporaneous selling activities of the remaining group members and serves as a measure of group-wide fluctuations in members' availability and workload that are common to all group members. We use the median rather than the mean to reduce the influence of outliers and to prevent unusually high or low activity by a single member from disproportionately affecting the measure. Conditional on this control and the other covariates in the model, it is unlikely that laggard active selling days affect group performance through channels other than laggard performance itself. In summary, laggard active selling days is likely to satisfy both the relevance criterion and the exclusion restriction and should thus constitute a valid instrument for laggard performance.

5.4 Model Specification

We estimate our model in two specifications. For the first specification, we estimate the group performance as a function of laggard performance using two-way fixed effects:

$$ln(Y_{it}) = \beta X_{it-1} + \delta_i + \lambda_t + \mu_{it}, \quad \mu_{it} \sim i.i.d.(0, \sigma^2) \tag{8}$$

where group performance Y_{it}, is the team average revenue for group i in month t. Laggard's performance X_{it-1} is the sales revenue of the laggard in group i at month $t-1$. δ_i are group fixed effects, λ_t capture the monthly fixed effects.

In the second specification, we use 2SLS employing laggard active selling days as an instrument. The first-stage regression and second-stage regression equations are:

$$X_{it-1} = \pi_0 + \pi_1 Z_{it-1} + \pi_2 W_{it-1} + \delta_i + \lambda_t + \mu_{it} \tag{9}$$

$$ln(Y_{it}) = \beta_1 \widehat{X_{it-1}} + \beta_2 W_{it-1} + \delta_i + \lambda_t + \mu_{it} \tag{10}$$

In the first stage of the 2SLS estimation, laggard performance X_{it-1} is modeled as a function of laggard active selling days, Z_{it-1}, for group i in month $t-1$. The vector W_{it-1} captures the active selling days of the remaining group members. The specification further includes group fixed effects δ_i and time fixed effects λ_t. We use the first-stage estimates to obtain the predicted value of laggard performance.

In the second stage, represented by equation (10), we regress group performance Y_{it} on predicted laggard performance, $\widehat{X}_{it-1}$. The specification again includes group fixed effects δ_i, time fixed effects λ_t, and W_{it-1}. Including W_{it-1} in both stages ensures that identification is driven by variation in the laggard's active selling days beyond contemporaneous group-wide variation in selling activity.

6 Results

6.1 Main effects of laggard performance on group performance

We present the results in Table 3. All independent variables are standardized. Across all models, the coefficient of laggard performance in month $t-1$ is positive and statistically significant in predicting group performance in month t, providing support for Hypothesis 1. Specifically, a one-standard-deviation increase in laggard performance is associated with an increase in group performance ranging from approximately 19.5% in the pooled OLS specification (Model 1; $\exp(0.178)-1$, $p<0.001$) to approximately 4.8% in the specification that includes both group fixed effects and month fixed effects (Model 3; $\exp(0.047)-1$, $p<0.001$). In the 2SLS specification (Model 4), the coefficient on instrumented laggard performance is also positive and statistically significant. The estimate implies that a one-standard-deviation increase in laggard performance increases group performance by approximately 3.6% ($\exp(0.035)-1$, $p<0.001$).

6.2 Robustness check

6.2.1 Lagged dependent variable

To ensure the robustness of our results and to address potential bias due to carryover effects in group performance, we incorporate a lagged measure of group performance into our model. In this analysis, we regress group performance in t on group performance in *t-1*, along with our main predictor, laggard performance. In addition, we include salesperson and month fixed effects to control for unobserved heterogeneity. Previous literature suggests that the incorporation of a lagged dependent variable along with fixed effects can bias the estimation, which is known as the "Nickell bias" (Nickell, 1981). This is due to the violation of the exogeneity assumption $\mathbb{E}[\mu_{it} \mid Y_{it}] = 0$, in equation 1. Specifically, lagged group performance carries the group performance information as well as the error term from the past month, which might be correlated with μ_{it} and the group fixed effect for the month t. However,

Table 3: Main Effects of Last-Place Aversion on Group Performance

	DV: ln(Group Performance)			
	(1) Estimates	(2) Estimates	(3) Estimates	(4) Estimates
Constant	9.859***			
	(0.007)			
Laggard performance $t-1$	0.178***	0.066***	0.047***	
	(0.007)	(0.017)	(0.004)	
Laggard performance $t-1$ (fitted)				0.035***
				(0.010)
Active selling days of group peers $t-1$				0.075***
				(0.015)
Observations	3,290	3,290	3,290	3,290
Group FE	No	Yes	Yes	Yes
Month FE	No	No	Yes	Yes
Clustered SE	No	Yes	Yes	Yes
R^2	0.184	0.647	0.835	0.839

Note: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$

Model 1 is the pooled OLS estimation. Model 2 includes group fixed effects. Model 3 includes both group and month fixed effects. Model 4 is the second stage of the 2SLS estimation. Standard errors are clustered by group and reported in parentheses. R^2 is based on the total variation explained by the full model.

as our data spans over 95 months, the Nickell Bias diminishes, given that the large number of time periods provides sufficient information to isolate the fixed effects from lagged group performance (Bond, 2002; Vergeer and Kleinknecht, 2012; Nickell and Saleheen, 2015).

Table 4: Lagged Dependent Variable Estimation

	DV: ln(Group Performance)	
	(1) Estimates	(2) Estimates
ln(Group performance $t-1$)	0.180***	0.168***
	(0.017)	(0.017)
Laggard performance $t-1$		0.017**
		(0.005)
Observations	3,290	3,290
Group FE	Yes	Yes
Month FE	Yes	Yes
Clustered SE	Yes	Yes
R^2	0.864	0.865

Note: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$

Standard errors are clustered by group and reported in parentheses. Group performance at $t-1$ denotes the lagged value of the dependent variable.

We present the results of the lagged dependent variable estimation in Table 4. Lagged group performance has a positive effect on group performance ($\beta = .180, p < .01$). Importantly, after taking this effect into account, the impact of laggard performance on group performance remains significant ($\beta = .017, p < .01$). Therefore, the lagged dependent variable estimation supports the positive effect of laggard performance on group performance.

6.2.2 Exclusion restriction of the instrument

We justified the relevance of our instrumental variable through F-statistic testing in the first stage of the 2SLS procedure (see results in Table 5). In relation to the exclusion restriction assumption, we rely on our institutional understanding of industrial selling in general and our empirical context in particular, positing that the instrument impacts group performance solely via its effect on laggard performance.

Table 5: First Stage Regression of 2SLS

DV: Laggard Performance $t-1$	
Instrument	
Laggard active selling days	0.583***
	(0.101)
Observations	3,290
First-stage F-statistic	2,361.8
Group FE	Yes
Month FE	Yes
Clustered SE	Yes
R^2	0.646

Note: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$
Standard errors are clustered by group and reported in parentheses. The reported coefficient is standardized.

To complement this reasoning and to increase confidence in the instrument's validity, we conduct a falsification exercise (Kiviet, 2020; Eggers et al., 2024): we include laggard active selling days *t–1* (our instrument) directly in the primary outcome equation (equation 1), alongside our main predictor, the full set of fixed effects, and *active selling days of group peers* as the control. If the instrument had a direct effect on group performance, we would expect its coefficient to remain significant in this specification.

We report the results of this regression in Table 6. The inclusion of laggard active selling days (Model 2) does not affect group performance ($\beta = -.004, p > .05$) and R^2 is unchanged across two models ($R^2 = .835$). These results remain consistent after we add *active selling days of group peers* as a control in Model 3. This provides indirect evidence consistent with our identifying assumption and alleviates concerns that the instrument influences group performance through channels other than laggard performance.

Table 6: Robustness Test for the Exclusion of the Instrument

	DV: ln(Group Performance)		
	(1) Estimates	(2) Estimates	(3) Estimates
Laggard performance $t-1$	0.047^{***}	0.050^{***}	0.040^{***}
	(0.004)	(0.007)	(0.005)
Laggard active selling days $t-1$		-0.004	-0.006
		(0.008)	(0.007)
Active selling days of group peers $t-1$			0.073^{***}
			(0.015)
Observations	3,290	3,290	3,290
Group FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes
R^2	0.835	0.835	0.839
Note:			$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$

Model 1 predicts group performance using laggard performance in month $t-1$. Model 2 adds the instrument, laggard active selling days, to the specification. Model 3 additionally includes the active selling days of group peers as a control, consistent with the IV specification. Standard errors are clustered by group and reported in parentheses.

6.3 Contingent effects of the last place aversion

To gain a more refined understanding of the conditions under which laggards might have more or less impact on group performance, we examined three potential moderators. These moderators include (a) laggard position change: whether the individual occupying the last-place position within a group changed or not since the previous month, (b) performance gap: the performance difference between the laggard and the group member who ranks just above them (i.e., second to last), and (c) magnitude of performance disparity: the performance difference between the laggard and the rest of their group members. In the following section, we elaborate on the measurement of these variables, followed by the estimation approach and model specification.

6.3.1 Measurement of moderating variables

Laggard position change indicates whether a different salesperson occupied last place relative to the preceding month. We measure it through a dummy variable that indicates whether the laggard was also in the last place in the previous month. Specifically, if a laggard for month *t-1* remains the same individual in month *t-2*, indicating a consistent bottom-rank salesperson over two consecutive months, we coded the variable as 0. Conversely, if there is a change in the laggard between month *t-1* and *t-2*, meaning two different salespeople occupied the last rank in these successive months, we coded the variable as 1.

Performance gap indicates the performance difference between the laggard and the group member who ranks just above them. This variable quantifies the distance in terms of the performance between the laggard and the penultimate peer. To calculate this moderator, we first identify the salesperson holding the penultimate position within the group in *t-1*. Following this, we subtract the laggard's revenue from the penultimate salesperson's revenue. This measure captures the extent of improvement the laggard needs to achieve in order to catch up or surpass the performance of the group member ranked immediately above them.

Magnitude of performance disparity depicts the extent to which the laggard performance deviates from the collective performance of their group members. We calculate it as the root mean square to quantify the magnitude of the difference between the laggard's revenue and the revenue of other group members in month $t-1$ (Wagner et al., 1984). The specific calculation is given by the following formula (11), where X_{it-1} is the laggard's revenue in month $t-1$ in group i, J_{it-1} is the revenue for a group member j from group i in the same period, and N_i is the total number of members for group i (excluding the laggard).

$$\text{Performance disparity} = \sqrt{\frac{1}{N_i}\sum_{i=1}^{N_i}(x_{t-1} - j_{t-1})^2} \tag{11}$$

6.3.2 Estimation approach

To model the effects of our moderating variables on the relationship between the laggard and group performance, we estimate group performance as a function of both the laggard performance, our moderators, and the interactions between the laggard performance and each of these moderating variables.

$$\ln(Y_{it}) = \beta_1 X_{it-1} + \sum_{k\in K}\{\beta_k m_k + \beta'_k(X_{it-1}m_{kt-1})\} + \delta_i + \lambda_t + \varepsilon_{it} \tag{12}$$

In this model, Y_{it} denotes the performance for group i at month t, X_{it-1} indicates the laggard's performance in group i at month $t-1$. β_j represents the coefficients of main effects of the set of moderators m_k, where k indexes the elements of the set of moderators K. β'_k indicate coefficients for the interactive effects between the laggard performance and the moderators. δ_i and λ_t are fixed effects.

However, this approach presents an identification challenge, which is particularly relevant when the interaction terms contain the endogenous variable, in this case, laggard performance. While the 2SLS approach for instrumental variable estimation can partly address this endogeneity, this model does not correct the endogeneity that arises from the interaction terms, which eventually leads to biased and inconsistent coefficients.

To solve this issue, we adopt the control function approach which allows us to handle endogeneity that comes from both the main predictor and its interaction terms. This approach involves using two-stage residual inclusion estimation (2SRI), which handles endogeneity by including the estimated residuals from the first stage in the second stage equation (Wooldridge, 2015).

In the first stage, we regress the laggard performance on the instrumental variable. We then extract the predicted residual of the first-stage regression. Equation (13) and (14) present this estimation, where X_{it-1} is the performance for the laggard in group i at month $t-1$, Z_{it-1} is the instrumental variable, laggard active selling days month $t-1$. Meanwhile, α_i and λ_t are the fixed effects. $\widehat{\mu_{it-1}}$ is the predicted residuals.

$$X_{it-1} = \pi_0 + \pi_1 Z_{it-1} + W_{it-1} + \alpha_i + \lambda_t + \mu_{it-1} \tag{13}$$

$$\widehat{\mu_{it-1}} = X_{it-1} - \widehat{X_{it-1}} \tag{14}$$

We then add the predicted residual as an additional regressor to equation (12). Thereby, we adjust for the non-zero conditional expectation of the error term, as evidenced by equation (13), where $\mathbb{E}[\varepsilon_{it}|\mu_{it-1} \neq 0]$. Hence, we rewrite our primary model as follows:

$$\ln(Y_{it}) = \beta_1 X_{it-1} + \sum_{k \in K} \{\beta_k m_k + \beta_k' (X_{it-1} m_{kt-1})\} + W_{it-1} + \delta_i + \lambda_t + \theta \widehat{\mu}_{it-1} + \varepsilon_{it} \tag{15}$$

In this specification, the inclusion of the control function allows us to isolate the endogenous portion of laggard performance and its associated interaction terms. Given the validity of our instrument as tested in the previous analysis, the OLS regression of this specification should provide consistent estimates.

6.3.3 Results of contingent effects of the laggard on group performance

We report the results of this analysis in Table 7. Our results confirm that the laggard's performance in month t-1 has a statistically significant positive impact on group performance across all three models. In Model 2, we include the interactions between laggard performance in month t-1 and the three moderators. All moderator variables were standardized to enhance interpretability of the results. The main effects of the performance difference between the laggard and the second-to-last group member, and performance difference between the laggard and other group members, are statistically significant. Model 3 includes the results of the 2SRI estimation. The predicted residual is statistically significant (.059, $p<$ .001), suggesting the presence of endogeneity in our measure of laggard performance. After controlling for this endogeneity through the control function, we observe decreased standard errors across all coefficients. Our following intepretations build on this model.

The key focus of our analysis lies in the interaction effects reported in Model 3. First, we find a positive and statistically significant interaction between laggard performance in month $t-1$ and laggard position change in predicting group performance ($\beta = 0.003$, $p < 0.05$), providing support for H_{2a}. This result indicates that when a different salesperson occupies the last-place position between months $t-2$ and $t-1$, the effect of laggard performance on group performance becomes stronger, thereby contributing to the average revenue of group members.

Second, we find a negative and statistically significant interaction between laggard performance in month $t-1$ and the performance difference between the laggard and the second-to-last-ranked group member in predicting group performance ($\beta = -0.003$, $p < 0.01$). This result supports H_{2b}. The performance gap captures the revenue difference between the laggard and the penultimate-ranked salesperson. The negative coefficient suggests that a larger performance gap weakens the effect of laggard performance on group performance.

Third, our results indicate a negative and statistically significant interaction between laggard performance in month $t-1$ and the performance disparity between the laggard and other group members in predicting group performance ($\beta = -0.005$, $p < 0.001$), providing support for H_{2c}. Because this moderator captures the disparity between the laggard's revenue and the revenue of other group members, the results suggest that greater disparity attenuates

Table 7: Results of Contingent Effects of Last-Place Aversion

	DV: *ln(Group Performance)*		
	(1)	(2)	(3)
	Estimates	Estimates	Estimates
Laggard performance $t-1$	0.055^{***} (0.007)	0.052^{***} (0.007)	0.044^{***} (0.001)
Laggard position change	0.010 (0.006)	0.011 (0.006)	0.009^{***} (0.001)
Performance gap	0.021^{***} (0.005)	0.020^{***} (0.005)	0.019^{***} (0.001)
Magnitude of performance disparity	0.066^{***} (0.012)	0.066^{***} (0.012)	0.062^{***} (0.002)
Active selling days of group peers $t-1$			0.027^{***} (0.002)
Laggard performance $t-1$ × laggard position change		0.006 (0.007)	0.003^{*} (0.001)
Laggard performance $t-1$ × performance gap		−0.003 (0.003)	-0.003^{**} (0.001)
Laggard performance $t-1$ × magnitude of performance disparity		−0.004 (0.004)	-0.005^{***} (0.001)
Residual			0.059^{***} (0.001)
Observations	3,211	3,211	3,211
Control function	No	No	Yes
Group FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes
R^2	0.85	0.85	Not applicable

Notes: Model 1 estimates the main effects of laggard performance and the moderators. Model 2 adds the interaction terms. Model 3 reports the two-stage residual inclusion (2SRI) estimates. Standard errors are clustered by group and reported in parentheses. *Residual* is the predicted residual from the first-stage estimation. R^2 is not reported for Model 3 because the model is estimated using a control function, for which the conventional R^2 statistic is not meaningful. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

the effect of laggard performance on group performance.

We present visualizations of these interaction effects in Figure 1.

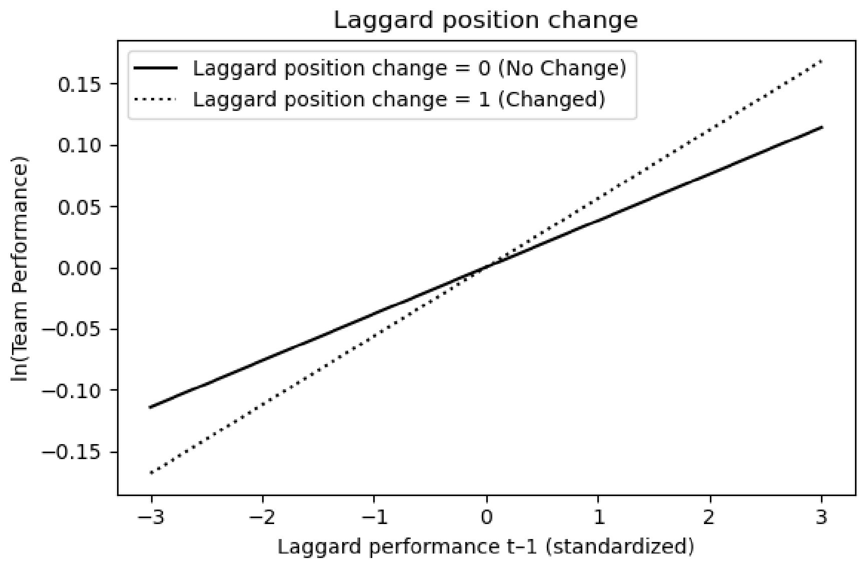

Figure 1: Effect of Laggard Position Change

Second, we find a significant and negative interactive effect between laggard performance in month $t-1$ and the performance gap on group performance ($\beta = -.005, p < .01$). The performance gap captures the revenue difference between the laggard and the penultimate-ranked salesperson. The negative coefficient suggests that an increase in the performance gap by one standard deviation weakens the effects of laggard performance on the group, leading to an additional 0.5% (= exp(.005) − 1) decrease in group performance.

Third, our results also suggest a significant and negative interaction effect between the laggard's performance in month $t-1$ and the magnitude of performance disparity on group performance ($\beta = -.004, p < .01$). As this moderator captures the magnitude of the difference between the laggard's revenue and the revenue of other group members, the results suggest that when the magnitude of performance disparity increases by one standard deviation, this attenuates the effect of the laggard on group performance, leading to an additional 0.4% (= exp(.004) − 1) decrease in group performance.

As both the performance gap and the magnitude of performance disparity are continuous variables, we demonstrate their interaction effects through a three-dimensional surface plot in Figure 2.

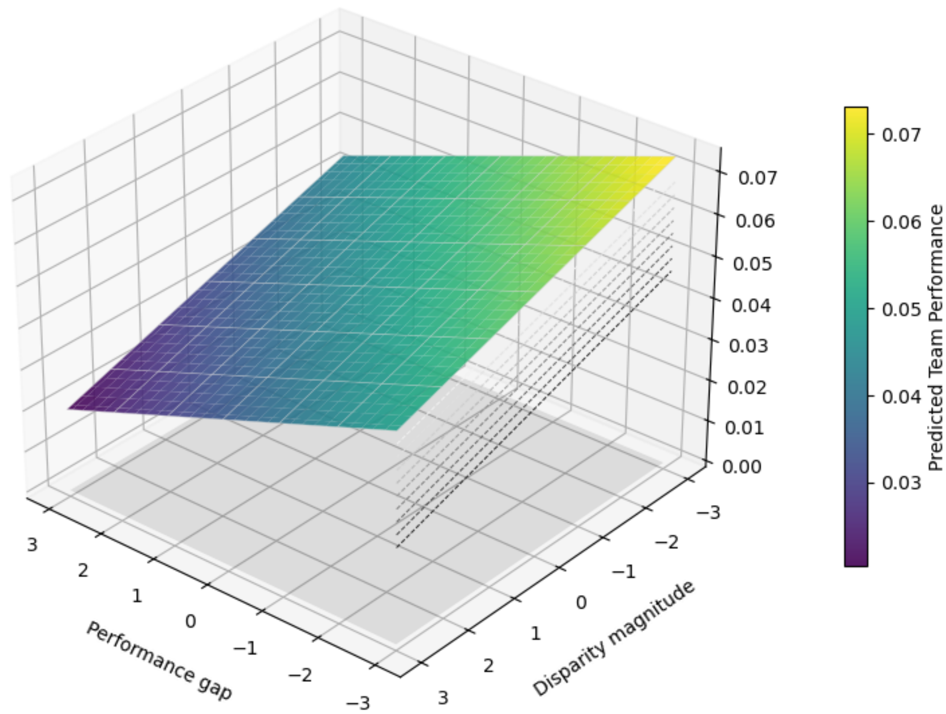

Figure 2: Visualization of the Interaction Effects.

6.4 Additional analysis

While our previous analysis indicates that the performance increase of a laggard positively impacts the average performance of their group, it does not capture the laggard's influence on individual group members occupying different ranks within the group. To pinpoint the main factors that contribute to group performance due to last place aversion, we further analyzed how the performance enhancement of a laggard affects salespeople at different performance ranks: those at the penultimate rank, those occupying the middle rank, and those in the top position.

We present the results of this analysis in Table 8. Models 1, 2, and 3 show the effect of laggard performance on the performance of the salespeople ranked at those three positions within their group. The results suggest that a one-standard-deviation increase in laggard performance leads to a 37.6% (= exp(.319) − 1) ($p < .01$) increase in the performance of the salesperson ranked second-lowest. Interestingly, the results in Model 2 suggest that the effect of laggard performance may extend to the salesperson ranked in the middle of the group, leading to a 3.36% (= exp(.033) − 1) ($p < .01$) performance increase. The decrease of the coefficient compared to Model 1 shows the diminishing influence of laggard performance as a focal salesperson's rank improves. Indeed, Model 3 suggests no significant relationship between the laggard's performance and that of the top-ranked group member, further underscoring that the influence of the laggard is most potent on group members who are ranked closer to them.

Table 8: Additional Analysis: Effects of Laggard Performance on Salespeople at Different Ranks

	Salesperson Performance at Respective Ranks		
	(1) Penultimate-ranked	(2) Median-ranked	(3) Top-ranked
Laggard Performance t-1	0.319***	0.033***	0.006
	(0.034)	(0.007)	(0.005)
Observations	3,215	3,245	3,245
group FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes
R^2	0.167	0.691	0.788

Note: *p<0.05; **p<0.01; ***p<0.001.
Dependent variables represent salesperson performance based on their rank within the group. Model 1 predicts the performance of the penultimate-ranked salesperson; Model 2 predicts the performance of the median-ranked salesperson; and Model 3 predicts the performance of the top-ranked salesperson. Standard errors are clustered at the group level.

7 Discussion and conclusions

Social comparisons are widely recognized as a key motivator for individuals to enhance their efforts and achieve their potential, making them a critical area of study in workplace settings where performance rankings are commonly adopted by organizations. In this research, we explore comparisons with individuals ranked at the bottom of a group's performance hierarchy, which we refer to as *laggards*. We examine whether group members exhibit an aversion to falling into last place, manifested in changes in the performance of the group, and individuals within it. Specifically, we focus on how the performance of laggards influences the performance outcomes of other group members.

Our results provide three key insights: First, improvements in laggard performance enhance the performance of other group members. Second, this effect differs between groups and depends on how salient it is to other group members that the individual being the laggard can change and the magnitude of performance differences to the laggard. Third, the effect of laggard performance improvements is more pronounced for individuals ranked just above them compared to those in middle or top ranks within the group.

This study is the first to investigate last place aversion in the context of performance rankings in the workplace, providing novel empirical evidence on how changes in laggards' performance are associated with subsequent work performance of their group members. Our findings contribute to the literature in several ways. First, we highlight that individuals exposed to performance-based rankings behave in ways which imply heightened sensitivity in downward comparisons, and specifically comparisons with those at the bottom of the hierarchy (Brown et al., 2007; Vriend et al., 2016). This insight complements existing research that has predominantly emphasized upward comparisons (Collins, 1996; Garcia and Tor, 2007; Obloj and Zenger, 2017; Waltré et al., 2023). Additionally, while prior studies have primarily examined social comparison effects in dyadic settings involving only two or three individuals and often relying on laboratory experimental methods (Shechter and Hardisty, 2020), our research extends this understanding by investigating these effects in real-world workplace groups. By accounting for group characteristics, our study demonstrates the interplay between laggards' performance improvements and the responses of their group members, offering novel insights into how these dynamics influence overall group performance.

Second, our study advances the understanding of last place aversion, a phenomenon previously explored in broader social contexts such as income distribution (Kuziemko et al., 2014)and queuing (Buell, 2021), by situating it in workplace group settings. While prior workplace research has examined the effects on individuals who themselves occupy the bottom place in rankings (Song et al., 2018; Buell, 2021), we find evidence indicative of *non-laggard* group members' aversion to falling into last place. Thus, we reveal a broader aversion to the last place within the group, as individuals strive to both leave and to avoid falling into this position. This extends previous findings by demonstrating that last place aversion is not limited to the individuals at the bottom (Gill et al., 2019) or those immediately adjacent to them (Kuziemko et al., 2014; Shechter and Hardisty, 2020)

Third, we uncover a novel influence on group performance. Specifically, we show that the improvement in a laggard's performance motivates other group members to improve their own performance, resulting in increased overall group performance. This finding highlights the spillover effects (Stan and Vermeulen, 2013) of laggards' performance improvements on group performance, extending existing studies which focus primarily on the self-motivated outcome of laggards (Gill et al., 2019; Buell, 2021). Additionally, we identify three group characteristics as key contingencies that moderate the relationship between laggards' performance and changes in their peers' performance. These contingencies indicate that the salience of falling into last place and the proximity to last place enhance last place aversion effects. Future research can extend our understanding of differences in last place aversion effects by exploring how rank dynamics in a more general sense interact with individual motivations in group settings.

Our findings offer valuable insights for managers seeking to enhance group performance through strategic interventions in rank-based environments. First, the evidence that laggards' performance improvements enhance broader group performance suggests that organizations should reconsider how they manage and support low-performing employees. Rather than marginalizing or removing laggards (Megaw, 2025), managers should focus on providing targeted resources and interventions to help them improve. For example, offering personalized coaching, skill development opportunities, or incentives tailored to laggards' specific challenges could not only elevate their individual performance but also create positive spillover effects for the entire group. By fostering a supportive environment that encourages laggards' growth (LePine and Van Dyne, 2001), managers can unlock motivation within the group, leading to overall performance gains.

Furthermore, greater transparency in performance rankings and recognition of progress may amplify the positive effects of laggard performance improvements. Our findings suggest that making group members aware of a laggard's progress can heighten the motivational impact of last place aversion across the hierarchy. Managers can leverage this insight by emphasizing laggards' efforts and achievements in group communications, such as during performance reviews or group meetings. Highlighting a narrowing performance gap between laggards and their peers can create competition, prompting group members to strive for higher performance levels. However, care should be taken to frame such communications positively to avoid undermining group cohesion or fostering counterproductive rivalries (Ahearne et al., 2025).

Finally, our study documents field evidence of performance dynamics consistent with last place aversion operating through downward comparisons with the laggard. In addition, future work could also extend our setting along several organizational dimensions. One avenue is to examine whether similar laggard-driven spillovers arise in groups with different task structures, alternative incentive schemes, or lower transparency of performance rankings (Friebel et al., 2017; Gjedrem and Kvaløy, 2020; Heursen, 2023). Such studies would help to assess the external validity of our results and contribute to a more general theory of how downward comparisons with the last place group member shape individual and collective performance in organizations.

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Appendix

A Model details and proofs

The group member chooses effort e_i to maximize expected utility:

$$U_i(e_i; y_L) = b(\alpha_i + \theta e_i) - \frac{\kappa}{2} e_i^2 - \psi \Phi \left(\frac{y_L - \alpha_i - \theta e_i}{\sigma} \right). \tag{16}$$

Define

$$z_i = \frac{y_L - \alpha_i - \theta e_i}{\sigma}. \tag{17}$$

The first derivative of expected utility with respect to effort is

$$\frac{\partial U_i}{\partial e_i} = b\theta - \kappa e_i + \frac{\psi \theta}{\sigma} \phi(z_i), \tag{18}$$

where $\phi(\cdot)$ denotes the standard normal density function.

An interior optimum e_i^* therefore satisfies

$$F(e_i, y_L) \equiv b\theta - \kappa e_i + \frac{\psi \theta}{\sigma} \phi(z_i) = 0. \tag{19}$$

Based on the implicit function theorem,

$$\frac{\partial e_i^*}{\partial y_L} = -\frac{F_{y_L}}{F_e}. \tag{20}$$

We first calculate F_{y_L}. Because

$$\phi'(z) = -z\phi(z) \tag{21}$$

and

$$\frac{\partial z_i}{\partial y_L} = \frac{1}{\sigma}, \tag{22}$$

we obtain

$$\begin{aligned} F_{y_L} &= \frac{\psi \theta}{\sigma} \phi'(z_i) \frac{\partial z_i}{\partial y_L} \\ &= \frac{\psi \theta}{\sigma} \phi'(z_i) \frac{1}{\sigma} \\ &= -\frac{\psi \theta}{\sigma^2} z_i \phi(z_i). \end{aligned} \tag{23}$$

For a non-laggard member above the laggard's performance,

$$\alpha_i + \theta e_i^* > y_L, \tag{24}$$

which implies

$$z_i < 0. \tag{25}$$

Since $\psi > 0$, $\theta > 0$, $\sigma > 0$, and $\phi(z_i) > 0$, it follows that

$$F_{y_L} > 0. \tag{26}$$

We next calculate F_e. Because

$$\frac{\partial z_i}{\partial e_i} = -\frac{\theta}{\sigma}, \tag{27}$$

we have

$$\begin{aligned} F_e &= -\kappa + \frac{\psi\theta}{\sigma}\phi'(z_i)\frac{\partial z_i}{\partial e_i} \\ &= -\kappa + \frac{\psi\theta}{\sigma}\phi'(z_i)\left(-\frac{\theta}{\sigma}\right). \end{aligned} \tag{28}$$

Using $\phi'(z_i) = -z_i\phi(z_i)$ gives

$$F_e = -\kappa + \frac{\psi\theta^2}{\sigma^2} z_i\phi(z_i). \tag{29}$$

Because $z_i < 0$,

$$\frac{\psi\theta^2}{\sigma^2} z_i\phi(z_i) < 0. \tag{30}$$

Since $\kappa > 0$, it follows that

$$F_e < 0. \tag{31}$$

Substituting equations (23) and (29) into equation (20) yields

$$\frac{\partial e_i^*}{\partial y_L} = \frac{-\psi\theta z_i\phi(z_i)/\sigma^2}{\kappa - \psi\theta^2 z_i\phi(z_i)/\sigma^2}. \tag{32}$$

For $z_i < 0$, the numerator in equation (32) is positive. The denominator is also positive because

$$-\psi\theta^2 z_i\phi(z_i)/\sigma^2 > 0. \tag{33}$$

Therefore,

$$\frac{\partial e_i^*}{\partial y_L} > 0. \tag{34}$$

An increase in the laggard's performance therefore raises the optimal effort of a non-laggard member whose expected performance lies above that laggard.

Because expected performance is increasing in effort,

$$\frac{\partial E[y_i]}{\partial e_i} = \theta > 0, \tag{35}$$

the increase in effort raises expected performance.

Let average group performance excluding the laggard be

$$Y^{-L} = \frac{1}{n-1} \sum_{i \neq L} y_i. \tag{36}$$

If an increase in y_L raises the optimal effort of at least some non-laggard group members, then

$$\frac{\partial E[Y^{-L}]}{\partial y_L} = \frac{\theta}{n-1} \sum_{i \neq L} \frac{\partial e_i^*}{\partial y_L} > 0. \tag{37}$$

Hence, an improvement in laggard performance increases subsequent group performance through its effect on the effort choices of non-laggard group members.

Table Appendix B1: Relevant Literature on Rank-based Performance Comparison

Authors	**Core Topic**	**Unit of Analysis**	**Theoretical Lens**	**Outcome Variable**	**Key Findings**	**Entire group?**	**Number of Moderators**	**Downward comparison?**	**Last place?**	**Time frame**
Obloj, Zenger (2017)	Organizational structure and productivity in response to social comparisons	Individual, Field Data	Social comparison theory	Individual productivity	Proximity to higher performers reduces productivity due to envy	No	0	No	No	2 months
Garcia, Tor (2007)	Task and scale comparisons affecting competitive behavior	Lab experiments with decision-making scenarios	Social comparison and competition theory	Competitive intensity	Competition intensifies when rivals are closer to meaningful standards	No	1	No	Yes	Not considered
Gartenberg, Wulf (2017)	Social comparison and compensation in multi-business firms	270 firms over 14 years	Social comparison and agency theory	Pay-performance sensitivity	Social comparison reduces pay-performance sensitivity, particularly through peer comparisons	No	0	No	No	14 years
Ahearne et al. (2025)	Performance ranking information impact on salespeople	Salespeople from 170 firms across 83 countries	Social comparison theory	Sales performance and turnover	Identifiable rankings improve quota attainment and reduce turnover; anonymized rankings increase turnover despite boosting performance	Yes	2	No	No	24 months
Kruger (1999)	Below-average effect in ability judgments	Individual, Controlled experiment	Anchoring and social comparison theory	Self-assessed comparative ranking	People rate themselves as above average in easy tasks but below average in challenging ones due to egocentric judgment	Yes	2	No	No	Not considered
Vriend, Jordan, Janssen (2016)	Effect of rankings on unethical behavior	Individual, Scenario-based experiments	Ranking theory	Unethical intentions	Higher-ranked individuals exhibit more unethical intentions than intermediate or lower ranks	No	2	Yes	Yes	Not considered

Table Appendix B2: Selected Literature on Last Place Aversion

Authors	**Core Topic**	**Unit of Analysis**	**Theoretical Lens**	**Outcome Variable**	**Key Findings**	**Non-last Individuals?**	**group Aggregated Outcome?**	**Job Performance?**	**Last Person Change?**	**Field Data?**
Gill et al. (2019)	Organization with a relative-performance evaluation system	Individual, Lab Experiment	Individual status seeking	Individual effort provision	Individuals exert the highest effort when they are in first or last place	No	No	No	No	No
Buell (2021)	Customers in queue settings	Individual, Field and Online Experiments	Last place aversion	Probability of switching queues	Individuals have a strong desire to avoid being in the last position in a ranking or queue	No	No	No	No	Yes
Song et al. (2018)	Relative Performance Feedback and the share of best practice in healthcare	Individual, Field Experiment	Social comparison	Individual productivity	Public relative performance feedback, with sharing of best practices, improves worker productivity	No	No	Yes	No	Yes
LePine, van Dyne (2001)	Individual members and low-performing peers in organizations	Collective, Theoretical Development	Attributional theory	Helping behaviors toward low performers	Low performers' abilities and efforts influence the helping behaviors of their peers	No	Yes	No	No	No
Current Paper	Last person's performance change and its impact on group	Both collective and individual level	econometric models	Aggregated performance of laggard's group members	Improvement in the laggard's performance increases overall group performance	Yes	Yes	Yes	Yes	Yes